

Math 121: Commutative Algebra

Prishita Dharampal

Worksheet 7.2: Integral Extension Pt. II

Throughout this course, “ring” means *commutative* ring with unity and all ring homomorphisms preserve unity. In the previous worksheet we introduced integral elements and integral extensions. We saw that integrality is the mechanism by which monic polynomial equations turn algebraic generation into module generation. The goal of this worksheet is to see what integrality does to prime ideals and to use this to make dimension computable for finitely generated algebras over a field.

Let $\phi : R \rightarrow S$ be a ring map. Recall that ϕ induces a continuous map

$$\mathrm{Spec}(S) \xrightarrow{\phi^\#} \mathrm{Spec}(R), \quad \mathfrak{q} \mapsto \phi^{-1}(\mathfrak{q}).$$

The central question of the first half of this worksheet is: *how much of $\mathrm{Spec}(R)$ is seen by $\mathrm{Spec}(S)$, and how do chains of prime ideals compare?* For integral extensions the answer is surprisingly rigid. Prime ideals downstairs have primes upstairs, chains can be lifted upward, and no two comparable upstairs primes can lie over the same downstairs prime. These facts are called *Lying Over*, *Going Up*, and *Incomparability*. A complementary statement, *Going Down*, is more delicate: it is true for flat maps, and for integral extensions of domains when the base is integrally closed.

In the second half of the worksheet we use these theorems to prove and apply *Noether Normalization*. The slogan is that every finitely generated algebra over a field is finite over a polynomial ring. Thus, after a finite map, an affine algebra behaves like affine space of some dimension. This is one of the main reasons polynomial rings control dimension theory.

Problem 1. A First Consequence for Maximal Ideals.

Let $\phi : R \rightarrow S$ be an injective integral ring map.

1. Suppose S is a field and R is a domain. Prove that R is a field. Hint: If $r \neq 0 \in R$, then $\phi(r)^{-1} \in S$ is integral over R ; write a monic equation for $\phi(r)^{-1}$ and multiply by a suitable power of $\phi(r)$.
2. Suppose S is a domain and R is a field. Prove that S is a field.
3. Let $\phi : R \rightarrow S$ be an integral ring map and let $\mathfrak{q} \in \text{Spec}(S)$ with contraction $\mathfrak{p} = \phi^{-1}(\mathfrak{q})$. Show that $S/\mathfrak{q}$ is integral over $R/\mathfrak{p}$.
4. Deduce that if $\mathfrak{q}$ is maximal, then $\mathfrak{p}$ is maximal.
5. Prove the converse: If $\mathfrak{p}$ is maximal, then $\mathfrak{q}$ is maximal. Hint: $S/\mathfrak{q}$ is an integral domain integral over the field $R/\mathfrak{p}$.
6. Conclude that under an integral ring map, a prime ideal upstairs is maximal if and only if its contraction downstairs is maximal.

Solution.

1. If $r \neq 0 \in R$ then $\phi(r)^{-1} \in S$ is integral over R with

$$(\phi(r)^{-1})^n + r_1(\phi(r)^{-1})^{n-1} + \cdots + r_n = 0$$

And,

$$\begin{aligned} 0 &= \phi(r)^n \left((\phi(r)^{-1})^n + r_1(\phi(r)^{-1})^{n-1} + \cdots + r_n \right) \\ 0 &= 1 + r_1\phi(r) + \cdots + r_n\phi(r)^n \\ \implies 1 &= \phi(r)(r_1 + \cdots + r_n\phi(r)^{n-1}) \end{aligned}$$

That is $\phi(r)$ has an inverse in $\phi(R)$. Since ϕ is injective this implies that r has an inverse in R . Since $\phi(r)$ was an arbitrary element this implies that every non-zero element in R has an inverse and R is a field.

2. We know that $s \neq 0 \in S$ is integral over R . That is, we have the minimal monic polynomial,

$$s^n + r_1s^{n-1} + \cdots + r_n = 0$$

Since R is a field we have,

$$\begin{aligned}
0 &= s^n + r_1 s^{n-1} + \cdots + r_n \\
r_n &= -(s^n + r_1 s^{n-1} + \cdots + r_{n-1} s) \\
-r_n &= s(s^{n-1} + r_1 s^{n-2} + \cdots + r_{n-1}) \\
1 &= \frac{-r_n}{-r_n} = s \left(\frac{1}{r_n} s^{n-1} + \frac{r_1}{r_n} s^{n-2} + \cdots + \frac{r_{n-1}}{r_n} \right)
\end{aligned}$$

That is s has an integral inverse in S . Since s was an arbitrary non-zero element this shows that S is a field.

3. Since S is integral over R for any $s \in S$ we have,

$$s^n + r_1 s^{n-1} + \cdots + r_n = 0$$

Then for $\bar{s} \in S/\mathfrak{q}$ we have,

$$\bar{s}^n + \bar{r}_1 \bar{s}^{n-1} + \cdots + \bar{r}_n = 0$$

But $\bar{r}_i = r_i + \mathfrak{q} \in S = r_i + \mathfrak{q} \cap R \in R = r_i + \mathfrak{p}$. Hence the coefficients $\bar{r}_i$ lie in $R/\mathfrak{p}$ and $\bar{s}$ is integral over $R/\mathfrak{p}$.

4. If $\mathfrak{q}$ is maximal, then $S/\mathfrak{q}$ is a field and then by part 1 $R/\mathfrak{f}\mathfrak{p}$ must be a field and hence $\mathfrak{p}$ must be maximal.
5. If $\mathfrak{p}$ is maximal, then $R/\mathfrak{p}$ is a field, and by part 2 $S/\mathfrak{q}$ is integral over $R/\mathfrak{p}$. Then for any $\bar{s} \neq 0 \in S/\mathfrak{q}$ we have the minimal monic polynomial,

$$\bar{s}^n + \bar{r}_1 \bar{s}^{n-1} + \cdots + \bar{r}_n = 0$$

And since $R/\mathfrak{p}$ is a field we have,

$$\begin{aligned}
\bar{r}_n &= -\bar{s}(\bar{s}^{n-1} + \bar{r}_1 \bar{s}^{n-2} + \cdots + \bar{r}_{n-1}) \\
1 &= \frac{\bar{r}_n}{\bar{r}_n} = -\bar{s} \left(\frac{1}{\bar{r}_n} \bar{s}^{n-1} + \frac{\bar{r}_1}{\bar{r}_n} \bar{s}^{n-2} + \cdots + \frac{\bar{r}_{n-1}}{\bar{r}_n} \right)
\end{aligned}$$

Thus $\bar{s}$ has an inverse in $S/\mathfrak{q}$, and hence $S/\mathfrak{q}$ is a field and $\mathfrak{q}$ is maximal.

6. Thus under an integral ring map $\mathfrak{q}$ is maximal if and only if $\mathfrak{p}$ is maximal.

Problem 2. Prime Ideals Near a Point.

Let $\phi : R \rightarrow S$ be a ring map and let $\mathfrak{p} \in \text{Spec}(R)$. Write $U = R \setminus \mathfrak{p}$ and $S_{\mathfrak{p}} = U^{-1}S$.

1. Prove that if $\mathfrak{q} \in \text{Spec}(S)$, then $\phi^{-1}(\mathfrak{q})$ is a prime ideal of R .
2. Prove that prime ideals of $S_{\mathfrak{p}}$ are in bijection with prime ideals $\mathfrak{q} \in \text{Spec}(S)$ such that $\mathfrak{q} \cap \phi(U) = \emptyset$. Equivalently, these are the primes $\mathfrak{q}$ such that $\phi^{-1}(\mathfrak{q}) \subseteq \mathfrak{p}$.
3. Suppose that $R \subseteq S$ is an injective ring map. Show that $S_{\mathfrak{p}} \neq 0$. Hint: If $1 = 0$ in $S_{\mathfrak{p}}$, then some element of $R \setminus \mathfrak{p}$ becomes zero in S .
4. Suppose that S is integral over R . Prove that $S_{\mathfrak{p}}$ is integral over $R_{\mathfrak{p}}$.
5. Let $\mathfrak{Q} \in \text{Spec}(S_{\mathfrak{p}})$ and let $\mathfrak{q} \subset S$ be the corresponding prime ideal. Show that if $\phi_{\mathfrak{p}}^{-1}(\mathfrak{Q}) = \mathfrak{p} R_{\mathfrak{p}}$, then $\phi^{-1}(\mathfrak{q}) = \mathfrak{p}$.

Solution.

1. If $\mathfrak{q} \in \text{Spec}(S)$ then $\mathfrak{q}$ is a prime ideal and we know that pre-images of prime ideals are prime, hence $\phi^{-1}(\mathfrak{q})$ is a prime ideal of R .
2. We have already proved this.
3. Assume for contradiction that $S_{\mathfrak{p}} = 0$. Then $\frac{0}{1} \sim \frac{1}{1} \implies \exists t \in \phi(R \setminus \mathfrak{p})$ such that $t(1 - 0) = 0 \implies t = 0 \implies t \in \ker(\phi)$ but ϕ is injective so t must be 0 and $0 \notin R \setminus \mathfrak{p}$. A contradiction. Hence, $S_{\mathfrak{p}} \neq 0$.
4. In Worksheet 7.1 Problem 17 that if S is integral over R and $U \subset R$ is multiplicatively closed, then $U^{-1}S$ is integral over $U^{-1}R$. Since $S_{\mathfrak{p}} = U^{-1}S$ with $U = R \setminus \mathfrak{p}$, we have $S_{\mathfrak{p}}$ is integral over $R_{\mathfrak{p}}$.
5. We have,

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ \downarrow & & \downarrow \\ R_{\mathfrak{p}} & \xrightarrow{\phi_{\mathfrak{p}}} & S_{\mathfrak{p}} \end{array}$$

Where

$$\phi_{\mathfrak{p}}^{-1}(\mathfrak{Q}) = \mathfrak{p} R_{\mathfrak{p}} = \mathfrak{Q} \cap R_{\mathfrak{p}}$$

And

$$\phi^{-1}(\mathfrak{q}) = \phi^{-1}(\mathfrak{Q} \cap S) \cap R = \phi_{\mathfrak{p}}^{-1}(\mathfrak{Q}) \cap R = \mathfrak{p} R_{\mathfrak{p}} \cap R = \mathfrak{p}$$

Problem 3. Lying Over.

Let $R \subseteq S$ be an integral extension of rings. The goal of this exercise is to prove the *Lying Over Theorem*: for every $\mathfrak{p} \in \text{Spec}(R)$, there exists $\mathfrak{q} \in \text{Spec}(S)$ such that $\mathfrak{q} \cap R = \mathfrak{p}$.

1. Fix $\mathfrak{p} \in \text{Spec}(R)$. Explain why $S_{\mathfrak{p}}$ is a nonzero integral extension of $R_{\mathfrak{p}}$.
2. Choose a maximal ideal $\mathfrak{N} \subset S_{\mathfrak{p}}$. Use the theorem from the previous worksheet saying that maximal ideals contract to maximal ideals under integral extensions to prove

$$\mathfrak{N} \cap R_{\mathfrak{p}} = \mathfrak{p} R_{\mathfrak{p}}.$$

3. Pull $\mathfrak{N}$ back to a prime ideal $\mathfrak{q} \subset S$ and prove that $\mathfrak{q} \cap R = \mathfrak{p}$.
4. State the correct version of Lying Over for a not-necessarily-injective integral map $\phi : R \rightarrow S$. Hint: Which prime ideals of R can possibly be contractions of prime ideals of S ?
5. Translate Lying Over into a statement about the image of the map $\phi^{\#} : \text{Spec}(S) \rightarrow \text{Spec}(R)$.

Solution.

1. In Worksheet 7.1 Problem 17 that if S is integral over R and $U \subset R$ is multiplicatively closed, then $U^{-1}S$ is integral over $U^{-1}R$. Since $S_{\mathfrak{p}} = U^{-1}S$ with $U = R \setminus \mathfrak{p}$, we have $S_{\mathfrak{p}}$ is integral over $R_{\mathfrak{p}}$.
2. Since maximal ideals contract to maximal ideals under integral extensions we know that $\phi_{\mathfrak{p}}^{-1}(\mathfrak{N})$ is a maximal ideal in $R_{\mathfrak{p}}$. Since there is only one maximal ideal in $R_{\mathfrak{p}}$, namely $\mathfrak{p} R_{\mathfrak{p}}$, we must have

$$\mathfrak{N} \cap R_{\mathfrak{p}} = \phi_{\mathfrak{p}}^{-1}(\mathfrak{N}) = \mathfrak{p} R_{\mathfrak{p}}$$

3. From Problem 2 part 5, we have that if $\mathfrak{q} \subset S$ corresponds to $\mathfrak{N}$ then

$$\phi^{-1}(\mathfrak{q}) = \mathfrak{q} \cap R = \mathfrak{p}$$

4. From the first isomorphism theorem we have $R/\ker(\phi) \cong \phi(R) \subseteq S$. Since ϕ is integral, prime ideals in S correspond to prime ideals in $R/\ker(\phi)$ which also correspond to prime ideals in R containing $\ker(\phi)$. So the Lying Over Theorem for not injective integral mps says that for every $\mathfrak{p} \in \text{Spec}(R)$ such that $\ker(\phi) \subseteq \mathfrak{p}$, there exists $\mathfrak{q} \in \text{Spec}(S)$ such that $\mathfrak{q} \cap R = \mathfrak{p}$.
5. The image of the map $\phi^{\#}$ is surjective onto $\text{Spec}(R/\ker(\phi))$, or the $\text{im}(\phi^{\#})$ is precisely the set of prime ideals in R containing the kernel of the map ϕ .

Problem 4. First Examples of Lying Over.

Let $\mathbb{K}$ be a field.

1. Show that $\mathbb{K}[t]$ is integral over $\mathbb{K}[t^2]$. For a prime ideal $\mathfrak{q} \subset \mathbb{K}[t]$, compute $\mathfrak{q} \cap \mathbb{K}[t^2]$ when $\mathfrak{q} = \langle t - a \rangle$.
2. Assume $\mathbb{K}$ is algebraically closed and $\text{char}(\mathbb{K}) \neq 2$. For $u = t^2$, describe the primes of $\mathbb{K}[t]$ lying over the maximal ideal $\langle u - a \rangle \subset \mathbb{K}[u]$ when $a = 0$ and when $a \neq 0$.
3. Show that $\mathbb{K}[t]$ is integral over $\mathbb{K}[t^2, t^3]$. Identify the prime of $\mathbb{K}[t]$ lying over the cusp point $\langle t^2, t^3 \rangle \subset \mathbb{K}[t^2, t^3]$.
4. Show that $\mathbb{Z}[i]$ is integral over $\mathbb{Z}$. Compute the primes of $\mathbb{Z}[i]$ lying over $\langle 2 \rangle$, $\langle 3 \rangle$, and $\langle 5 \rangle$ by factoring $x^2 + 1$ modulo 2, 3, and 5.
5. Let $R \subseteq S$ be integral and let $I \subset R$ be an ideal. Prove that the induced map

$$\{\mathfrak{q} \in \text{Spec}(S) \mid \mathfrak{q} \supseteq IS\} \longrightarrow \{\mathfrak{p} \in \text{Spec}(R) \mid \mathfrak{p} \supseteq I\}$$

is surjective.

Solution.

1. $\mathbb{K}[t]$ is module-finite over $\mathbb{K}[t^2]$ and hence is integral over $\mathbb{K}[t^2]$.

$$\mathfrak{q} \cap \mathbb{K}[t^2] = \{p(t^2) \in \mathbb{K}[t^2] \mid p(a^2) = 0\} = \langle t^2 - a^2 \rangle$$

2. Prime ideals that lie over the maximal ideal $\mathfrak{p} = \langle t^2 - a \rangle \subseteq \mathbb{K}[t^2]$ are maximal ideals $\mathfrak{q}$ in $\mathbb{K}[t]$ such that $\mathfrak{q} \cap \mathbb{K}[t^2] = \mathfrak{p}$. Maximal ideals in $\mathbb{K}[t]$ are of the form $\langle t - b \rangle$, and by part 1 we know that $\langle t - b \rangle$ lies over $\mathfrak{p}$ iff $b^2 = a$.

If $a = 0$,

$$\langle t - b \rangle \cap \mathbb{K}[t^2] = \langle t^2 \rangle \implies b = 0$$

So there is a single prime ideal $\langle t \rangle$ lying over $\langle t^2 \rangle$.

If $a \neq 0$, then $b = \pm\sqrt{a}$ since $\mathbb{K}$ is algebraically closed. Hence, two prime ideals lie over $\mathfrak{p}$, $\langle t - \sqrt{a} \rangle$ and $\langle t + \sqrt{a} \rangle$.

3. $\mathbb{K}[t]$ is module-finite over $\mathbb{K}[t^2, t^3]$ and hence is integral over $\mathbb{K}[t^2, t^3]$. Primes of $\mathbb{K}[t]$ are of the form $\langle t - b \rangle$ for $b \in \mathbb{K}$ and $\langle 0 \rangle$. If $\langle t - b \rangle$ lies over $\langle t^2, t^3 \rangle$ then

$$t^2, t^3 \in \langle t - b \rangle \implies b^2 = b^3 = 0 \implies b = 0$$

Thus, $\langle t \rangle$ lies over $\langle t^2, t^3 \rangle$.

4. $\mathbb{Z}[i]$ is module-finite over $\mathbb{Z}$ and hence is integral over $\mathbb{Z}$. We know $\mathbb{Z}[i] \cong \mathbb{Z}[x]/\langle x^2 + 1 \rangle$, and primes of $\mathbb{Z}[i]$ containing $\langle p \rangle$ correspond to prime ideals of $\mathbb{Z}[i]/\langle p \rangle$. Then,

$$\mathbb{Z}[i]/\langle p \rangle \cong \mathbb{F}_p[i] \cong \mathbb{F}_p[x]/\langle x^2 + 1 \rangle$$

so primes correspond to irreducible factors of $\langle x^2 + 1 \rangle$ in $\mathbb{F}_p[x]$ since they generate the ideals containing it. Therefore,

- $x^2 + 1 = (x + 1)^2 \pmod{2}$ so $\langle 1 + i \rangle$ lies over $\langle 2 \rangle$.
- $x^2 + 1$ is irreducible modulo 3 since $3 \pmod{4} \equiv 3$ and thus -1 is not a square, so $\langle 3 \rangle$ lies over $\langle 3 \rangle$.
- $x^2 + 1 = (x + 2)(x - 2) \pmod{5}$ so $\langle i + 2 \rangle$ and $\langle i - 2 \rangle$ lie over $\langle 5 \rangle$.

5. Let $\mathfrak{p} \in \text{Spec}(R)$ such that $I \subseteq \mathfrak{p}$. Then by laying over we know that there exists $\mathfrak{q}$ in $\text{Spec}(S)$ such that $\mathfrak{q} \cap R = \mathfrak{p}$. Since $\mathfrak{q}$ is an ideal and we have $I \subseteq \mathfrak{p} \subseteq \mathfrak{q}$ as sets, $IS \subseteq \mathfrak{q}$. Hence, proving surjectivity.

The next theorem explains how integral extensions interact with inclusions of prime ideals. Geometrically, if $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ in $\text{Spec}(R)$, then $\mathfrak{p}_1$ is a specialization of $\mathfrak{p}_0$. The theorem says that specializations can be lifted through an integral extension once the starting point upstairs has been chosen.

Theorem 0.1 (Going Up). *Let $R \subseteq S$ be an integral extension. Suppose*

$$\mathfrak{p}_0 \subseteq \mathfrak{p}_1$$

are prime ideals of R , and suppose $\mathfrak{q}_0 \in \text{Spec}(S)$ satisfies $\mathfrak{q}_0 \cap R = \mathfrak{p}_0$. Then there exists $\mathfrak{q}_1 \in \text{Spec}(S)$ such that

$$\mathfrak{q}_0 \subseteq \mathfrak{q}_1 \quad \text{and} \quad \mathfrak{q}_1 \cap R = \mathfrak{p}_1.$$

More generally, every chain of prime ideals downstairs can be lifted to a chain of prime ideals upstairs, after choosing a prime upstairs over the first prime in the chain.

Problem 5. Proof of Going Up.

Let $R \subseteq S$ be integral, let $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ be prime ideals of R , and let $\mathfrak{q}_0 \in \text{Spec}(S)$ lie over $\mathfrak{p}_0$.

1. Prove that $S/\mathfrak{q}_0$ is integral over $R/\mathfrak{p}_0$.
2. Show that $\mathfrak{p}_1/\mathfrak{p}_0$ is a prime ideal of $R/\mathfrak{p}_0$.
3. Apply Lying Over to the integral extension $R/\mathfrak{p}_0 \subseteq S/\mathfrak{q}_0$ to produce a prime ideal $\mathfrak{q}_1/\mathfrak{q}_0$ of $S/\mathfrak{q}_0$ lying over $\mathfrak{p}_1/\mathfrak{p}_0$.
4. Check that the corresponding prime ideal $\mathfrak{q}_1 \subset S$ satisfies $\mathfrak{q}_0 \subseteq \mathfrak{q}_1$ and $\mathfrak{q}_1 \cap R = \mathfrak{p}_1$.
5. Prove the chain version of Going Up by induction: if

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

is a chain in R and $\mathfrak{q}_0$ lies over $\mathfrak{p}_0$, then there is a chain

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n$$

in S with $\mathfrak{q}_i \cap R = \mathfrak{p}_i$ for all i .

Solution.

1. This follows from Problem 1 part 5.
2. We know that prime ideals of $R/\mathfrak{p}_0$ correspond to prime ideals of R containing $\mathfrak{p}_0$. Since $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$, $\overline{\mathfrak{p}_1} = \mathfrak{p}_1/\mathfrak{p}_0$ is a prime ideal of $R/\mathfrak{p}_0$.
3. We know that the map $R/\mathfrak{p}_0 \rightarrow S/\mathfrak{q}_0$ is an integral map. By laying over, we know there exists $\overline{\mathfrak{q}_1} \subseteq S/\mathfrak{q}_0$ that lies over $\overline{\mathfrak{p}_1}$. Since prime ideals of $S/\mathfrak{q}_0$ correspond to prime ideals of S containing $\mathfrak{q}_0$, $\overline{\mathfrak{q}_1} = \mathfrak{q}_1/\mathfrak{q}_0$.
4. Thus, $\mathfrak{q}_0 \subseteq \mathfrak{q}_1$. We know $\mathfrak{q}_1/\mathfrak{q}_0 \cap R/\mathfrak{p}_0 = \mathfrak{p}_1/\mathfrak{p}_0$. $\mathfrak{q}_1 \cap R$ is the pre-image of the quotient map $\pi : R \rightarrow R/\mathfrak{p}_0$, then,

$$\pi^{-1}((\mathfrak{q}_1/\mathfrak{q}_0) \cap (R/\mathfrak{p}_0)) = \pi^{-1}(\mathfrak{p}_1/\mathfrak{p}_0) = \mathfrak{p}_1$$

5. Assume for $m < n$ that if

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m$$

is a chain in R and $\mathfrak{q}_0$ lies over $\mathfrak{p}_0$, then there is a chain

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_m$$

in S with $q_i \cap R = p_i$ for all i .

Then if $p_m \subsetneq p_n$, then by parts 3 and 4, we know there exists $q_n \subseteq S$ such that $q_m \subset q_n$ and $q_n \cap R = p_n$. Note that the inclusion $q_m \subsetneq q_n$ is strict since the contractions correspond to p_m and p_n , and the inclusion $p_m \subsetneq p_n$ is strict.

Thus for any n , if

$$p_0 \subsetneq p_1 \subsetneq \cdots \subsetneq p_n$$

is a chain in R and q_0 lies over p_0 , then there is a corresponding chain

$$q_0 \subsetneq q_1 \subsetneq \cdots \subsetneq q_n$$

in S with $q_i \cap R = p_i$ for all i .

Problem 6. Incomparability.

Let $R \subseteq S$ be an integral extension. The goal of this exercise is to prove: if $\mathfrak{q} \subseteq \mathfrak{q}'$ are prime ideals of S and $\mathfrak{q} \cap R = \mathfrak{q}' \cap R$, then $\mathfrak{q} = \mathfrak{q}'$.

1. Set $\mathfrak{p} = \mathfrak{q} \cap R = \mathfrak{q}' \cap R$. Explain why $R/\mathfrak{p} \subseteq S/\mathfrak{q}$ is an integral extension of domains.
2. Let $\mathfrak{P} = \mathfrak{q}'/\mathfrak{q} \subset S/\mathfrak{q}$. Show that $\mathfrak{P} \cap (R/\mathfrak{p}) = \langle 0 \rangle$.
3. Suppose $0 \neq b \in \mathfrak{P}$. Choose a monic equation of minimal degree

$$b^n + a_1 b^{n-1} + \cdots + a_{n-1} b + a_n = 0$$

with $a_i \in R/\mathfrak{p}$. Prove that $a_n = 0$.

4. Use the fact that $S/\mathfrak{q}$ is a domain to get a monic equation for b of smaller degree, a contradiction.
5. Conclude Incomparability.

Solution.

1. This follows from Problem 1 part 5.

2. If $r + \mathfrak{q}, r' + \mathfrak{q} \in \mathfrak{q}'/\mathfrak{q}$ then $r - r' \in \mathfrak{q} \subseteq \mathfrak{q}'$, that is,

$$\mathfrak{P} \cap (R/\mathfrak{p}) = \{r + \mathfrak{p} \mid r + \mathfrak{q} \in \mathfrak{q}'/\mathfrak{q}\} = \{r + \mathfrak{p} \mid r \in \mathfrak{q}' \cap R\} = \{r + \mathfrak{p} \mid r \in \mathfrak{p}\} = \langle 0 \rangle$$

3. Since $b \in \mathfrak{P}$ we have $b^n + a_1 b^{n-1} + \cdots + a_{n-1} b \in \mathfrak{P}$ which implies that $-a_n \in \mathfrak{P}$. But since $a_n \in R/\mathfrak{p} \implies a_n \in \mathfrak{P} \cap (R/\mathfrak{p}) = \langle 0 \rangle \implies a_n = 0$.
4. We have,

$$b(b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}) = 0$$

Since $S/\mathfrak{q}$ is a domain and $b \neq 0$ we must have

$$b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1} = 0$$

Thus giving us a monic equation for b of smaller degree, which is a contradiction since we assumed minimality. Thus $b = 0$.

5. Part 4 shows us that $\mathfrak{P}$ has no non-zero elements and thus $\mathfrak{q}'/\mathfrak{q} = \langle 0 \rangle \implies \mathfrak{q}' = \mathfrak{q}$.

Problem 7. Dimension is Preserved by Integral Extensions.

Let $R \subseteq S$ be an integral extension of nonzero rings.

1. Let

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n$$

be a chain of prime ideals in S . Use Incomparability to show that

$$\mathfrak{q}_0 \cap R \subsetneq \mathfrak{q}_1 \cap R \subsetneq \cdots \subsetneq \mathfrak{q}_n \cap R$$

is a chain of prime ideals in R . Deduce that $\dim(S) \leq \dim(R)$.

2. Let

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

be a chain of prime ideals in R . Use Lying Over and Going Up to lift it to a chain of prime ideals in S . Deduce that $\dim(R) \leq \dim(S)$.

3. Conclude that $\dim(R) = \dim(S)$.

4. Use this theorem to compute the Krull dimension of each ring below.

(a) $\mathbb{K}[t^2, t^3]$.

(c) $\mathbb{K}[x^2, xy, y^2]$.

(b) $\mathbb{K}[x, y] / \langle y^2 - x^3 \rangle$.

(d) $\mathbb{Z}[i]$.

Solution.

1. Since the preimage of prime ideals is prime we know that each $\mathfrak{q}_i \cap R$ is prime, and by Incomparability all containments are strict. Then if

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n$$

is the longest chain of prime ideals in S , we get a chain of prime ideals in R that has the same length

$$\mathfrak{q}_0 \cap R \subsetneq \mathfrak{q}_1 \cap R \subsetneq \cdots \subsetneq \mathfrak{q}_n \cap R$$

Hence $\dim(S) \leq \dim(R)$.

2. If

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

is a chain of prime ideals in R . Then by laying over we know that there exist prime ideals

$$\mathfrak{q}_0, \mathfrak{q}_1, \dots, \mathfrak{q}_n$$

lying over each corresponding ideal in R . Moreover from going over we know that for each pair $\mathfrak{p}_i \subseteq \mathfrak{p}_{i+1}$ such that there exists a $\mathfrak{q}_i$ that lies over $\mathfrak{p}_i$, there exists an prime ideal $\mathfrak{q}_{i+1} \subseteq S$ that lies over $\mathfrak{p}_{i+1}$ and contains $\mathfrak{q}_i$. Thus we get a chain of prime ideals in S ,

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n$$

Since the chains are of equal length we have $\dim(S) \geq \dim(R)$.

3. Thus, $\dim(S) = \dim(R)$.

4. (a) Since $\mathbb{K}[t]$ is integral over $\mathbb{K}[t^2, t^3]$,

$$\dim(\mathbb{K}[t^2, t^3]) = \dim(\mathbb{K}[t]) = 1$$

(b) Since $\mathbb{K}[x, y] / \langle y^2 - x^3 \rangle$ is module-finite (and thus integral over) $\mathbb{K}[x]$,

$$\dim(\mathbb{K}[x, y] / \langle y^2 - x^3 \rangle) = \dim(\mathbb{K}[x]) = 1$$

(c) Since $\mathbb{K}[x, y]$ is integral over $\mathbb{K}[x^2, xy, y^2]$,

$$\dim(\mathbb{K}[x, y]) = \dim(\mathbb{K}[x^2, xy, y^2]) = 2$$

(d) Since $\mathbb{Z}[i]$ is integral over $\mathbb{Z}$,

$$\dim(\mathbb{Z}[i]) = \dim(\mathbb{Z}) = 1$$

Problem 8. Closed Sets Under Integral Maps.

Let $R \subseteq S$ be integral and let $\phi^\# : \text{Spec}(S) \rightarrow \text{Spec}(R)$ be the induced map.

1. Let $J \subseteq S$ be an ideal. Show that

$$\phi^\#(\mathbb{V}(J)) = \mathbb{V}(J \cap R).$$

Hint: One inclusion is direct. For the other, apply Lying Over to $R/(J \cap R) \subseteq S/J$ after checking the kernel carefully.

2. Deduce that $\phi^\#$ is a closed map.
3. If $R \subseteq S$ is finite and κ is algebraically closed, explain why the induced map on affine algebraic sets should be thought of as a finite-to-one map. Make this precise by showing that for each $\mathfrak{p} \in \text{Spec}(R)$, the fibre ring $\kappa(\mathfrak{p}) \otimes_R S$ is finite-dimensional over $\kappa(\mathfrak{p})$.

Solution.

1. $\phi^\#(\mathbb{V}(J)) \subseteq \mathbb{V}(J \cap R)$

Let $\mathfrak{p} \in \phi^\#(\mathbb{V}(J))$. Then there exists some $\mathfrak{q} \in \mathbb{V}(J)$ such that $\phi^\#(\mathfrak{q}) = \mathfrak{q} \cap R = \mathfrak{p}$. Since $J \subseteq \mathfrak{q}$ we must have $J \cap R \subseteq \mathfrak{q} \cap R = \mathfrak{p}$.

$$\phi^\#(\mathbb{V}(J)) \supseteq \mathbb{V}(J \cap R)$$

Let $\mathfrak{p} \in \mathbb{V}(J \cap R)$. Then by going up and laying over we have,

$$\begin{array}{ccc} J & \subseteq & \mathfrak{q} \\ \downarrow & & \downarrow \\ J \cap R & \subseteq & \mathfrak{p} \end{array}$$

Thus there exists an $\mathfrak{q} \subset S$ such that $\phi^\#(\mathfrak{q}) = \mathfrak{p}$.

2. Thus, $\phi^\#$ maps closed sets to closed sets and hence is a closed map.
3. If $R \subseteq S$ is finite and κ is algebraically closed, explain why the induced map on affine algebraic sets should be thought of as a finite-to-one map. Make this precise by showing that for each $\mathfrak{p} \in \text{Spec}(R)$, the fibre ring $\kappa(\mathfrak{p}) \otimes_R S$ is finite-dimensional over $\kappa(\mathfrak{p})$.

$$\begin{array}{ccc} R & \longrightarrow & \kappa(\mathfrak{p}) \\ \downarrow & & \downarrow \\ S & \longrightarrow & \kappa(\mathfrak{p}) \otimes_R S \end{array}$$

Note that every element of $\kappa(\mathfrak{p}) \otimes_R S$ is a finite sum of simple tensors $k \otimes s$. Since $s \in S$, we can write $s = \sum_i r_i s_i$ for $r_i \in R$. Then,

$$\begin{aligned}
 k \otimes s &= k \otimes \sum_i r_i s_i \\
 &= k \sum_i r_i \otimes \sum_i s_i \\
 &= \sum_i k r_i \otimes s_i \\
 &= \sum_i k r_i (1 \otimes s_i)
 \end{aligned}$$

Hence $\kappa(\mathfrak{p}) \otimes_R S$ is spanned by simple tensors $(1 \otimes s_1), \dots, (1 \otimes s_n)$ and thus is finite dimensional over $\kappa(\mathfrak{p})$.

We now turn to *Going Down*. For a ring map $R \rightarrow S$, Going Down means that chains can be lifted in the opposite direction: if $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ in R and a prime $\mathfrak{q}_1$ upstairs has already been chosen over $\mathfrak{p}_1$, then there should exist a prime $\mathfrak{q}_0 \subseteq \mathfrak{q}_1$ lying over $\mathfrak{p}_0$.

Going Down is not automatic for integral extensions. There are two especially important settings where it holds. The first is flatness, which is a module-theoretic condition. The second is the classical integral theorem: an integral extension of domains satisfies Going Down when the base is integrally closed. We will not prove these now, but will explore their consequences.

Theorem 0.2 (Flat Going Down). *Let $\phi : R \rightarrow S$ be a flat ring map. Suppose $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ are prime ideals of R and $\mathfrak{q}_1 \in \text{Spec}(S)$ satisfies $\phi^{-1}(\mathfrak{q}_1) = \mathfrak{p}_1$. Then there exists $\mathfrak{q}_0 \in \text{Spec}(S)$ such that*

$$\mathfrak{q}_0 \subseteq \mathfrak{q}_1 \quad \text{and} \quad \phi^{-1}(\mathfrak{q}_0) = \mathfrak{p}_0.$$

Theorem 0.3 (Going Down for Normal Domains). *Let $A \subseteq B$ be an integral extension of domains. If A is integrally closed in its fraction field, then $A \subseteq B$ satisfies Going Down.*

Problem 9. Unpacking Going Down.

Let $\phi : R \rightarrow S$ be a ring map.

1. Write the definition of Going Up for ϕ using a diagram of prime ideals.
2. Write the definition of Going Down for ϕ using a diagram of prime ideals.
3. Explain why Going Up is about lifting specializations and Going Down is about lifting generalizations.
4. Prove directly that a localization map $R \rightarrow U^{-1}R$ satisfies Going Down. Hint: Use the correspondence between primes of $U^{-1}R$ and primes of R disjoint from U .
5. Give an example of a flat map that is not integral and an integral map that is not flat.

Solution.

1. Going up says, given $\mathfrak{q}_0$ over $\mathfrak{p}_0$ and $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ we have

$$\begin{array}{ccc} \mathfrak{q}_0 & \overset{\subseteq}{\dashrightarrow} & \mathfrak{q}_1 \\ \cap R \downarrow & & \downarrow \cap R \\ \mathfrak{p}_0 & \xrightarrow{\subseteq} & \mathfrak{p}_1 \end{array}$$

2. Going down says, given $\mathfrak{q}_1$ over $\mathfrak{p}_1$ and $\mathfrak{p}_0 \subseteq \mathfrak{p}_1$ we have

$$\begin{array}{ccc} \mathfrak{q}_0 & \overset{\subseteq}{\dashleftarrow} & \mathfrak{q}_1 \\ \cap R \downarrow & & \downarrow \cap R \\ \mathfrak{p}_0 & \xrightarrow{\subseteq} & \mathfrak{p}_1 \end{array}$$

3. In $\text{Spec}(S)$ going up enables us to go from $\mathfrak{q}_0$ (over $\mathfrak{p}_0$) to $\mathfrak{q}_0 \subseteq \mathfrak{q}_1$, so $\mathfrak{q}_1 \in \mathbb{V}(\mathfrak{q}_0)$ and going down lets us find $\mathfrak{q}_0$ such that $\mathfrak{q}_1 \in \mathbb{V}(\mathfrak{q}_0)$.
4. In worksheet 6.1 we showed that a localized ring is a flat module and thus localization satisfies the flat going down theorem.
5. $\mathbb{K} \rightarrow \mathbb{K}[x]$ is flat but not integral and $\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}$ is integral but not flat.

Problem 10. Consequences of Going Down.

1. Let $R \rightarrow S$ be faithfully flat. Prove that $\text{Spec}(S) \rightarrow \text{Spec}(R)$ is surjective. Hint: For $\mathfrak{p} \in \text{Spec}(R)$, show that $\kappa(\mathfrak{p}) \otimes_R S \neq 0$.
2. Deduce that if $R \rightarrow S$ is faithfully flat, then every finite chain of primes in R can be lifted to a chain of primes in S . Conclude that $\dim(S) \geq \dim(R)$.
3. Apply this to $R \rightarrow R[x]$ to recover the inequality $\dim(R[x]) \geq \dim(R) + 1$ for nonzero R . Hint: if $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ is a chain in R , use the explicit chain $\mathfrak{p}_0 R[x] \subsetneq \cdots \subsetneq \mathfrak{p}_n R[x] \subsetneq \mathfrak{p}_n R[x] + \langle x \rangle$.
4. Let $A \subseteq B$ be an integral extension of domains with A integrally closed. Let $\mathfrak{q} \in \text{Spec}(B)$ and $\mathfrak{p} = \mathfrak{q} \cap A$. Use Going Up, Going Down for Normal Domains, and Incomparability to prove

$$\text{ht}(\mathfrak{q}) = \text{ht}(\mathfrak{p}).$$

5. Specialize the previous part to a finite extension $\mathbb{K}[y_1, \dots, y_d] \subseteq B$ where B is a domain. What does it say about heights of primes in B ?

Solution.

1. Since $R \rightarrow S$ is faithfully flat we know that $\kappa(\mathfrak{p}) \otimes_R S = 0 \iff \kappa(\mathfrak{p}) = 0$, as $\kappa(\mathfrak{p}) \neq 0$ for any prime ideal we know $\kappa(\mathfrak{p}) \otimes_R S \neq 0$. That is $\kappa(\mathfrak{p}) \otimes_R S$ has at least one prime ideal, let this ideal be $\bar{\mathfrak{q}}$. Since pre-images of prime ideals are prime, $\bar{\mathfrak{q}}$ gives us a prime ideal $\mathfrak{q} \in S$.

$$\begin{array}{ccc} S & \longrightarrow & \kappa(\mathfrak{p}) \otimes_R S \\ \uparrow & & \uparrow \\ R & \longrightarrow & \kappa(\mathfrak{p}). \end{array}$$

Since the diagram commutes, the two maps $R \rightarrow S \rightarrow \kappa(\mathfrak{p}) \otimes_R S$ and $R \rightarrow \kappa(\mathfrak{p}) \rightarrow \kappa(\mathfrak{p}) \otimes_R S$ must agree. Pulling back $\bar{\mathfrak{q}}$ along the bottom-right path gives us $(0) \in \text{Spec}(\kappa(\mathfrak{p}))$, and (0) contracts to $\mathfrak{p} \in \text{Spec}(R)$. Thus $\mathfrak{q}$ lies over $\mathfrak{p}$. Hence every $\mathfrak{p} \in \text{Spec}(R)$ is in the image and thus the map $\text{Spec}(S) \rightarrow \text{Spec}(R)$ is surjective.

2. Since $\text{Spec}(S) \rightarrow \text{Spec}(R)$ is surjective, given a longest prime chain in R ,

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

we can find a lift of $\mathfrak{p}_n, \mathfrak{q}_n \in \text{Spec}(S)$. Then by going down, we can inductively find the corresponding prime chain of the same length in S ,

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n$$

so $\dim(S) \geq \dim(R)$.

3. Let

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

be any chain in R and define the ideal lying over $\mathfrak{p}_n$ as $\mathfrak{q}_n = \mathfrak{p}_n R[x]$ since $\mathfrak{q} \cap R = \mathfrak{p}$ and

$$R[x] / \mathfrak{q}_n = R / \mathfrak{p}_n R[x] = (R / \mathfrak{p}_n)[x]$$

which is a domain. Then by part (2) we can inductively find a chain such that

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n = \mathfrak{p}_n R[x]$$

Note that $\mathfrak{p}_n R[x] \subsetneq \mathfrak{p}_n R[x] + \langle x \rangle$ since $x \notin \mathfrak{p}_n R[x]$ because every element of $\mathfrak{p}_n R[x]$ has its coefficients in $\mathfrak{p}_n$ but x has leading coefficient 1.

So we have a chain of length $n + 1$ in S

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n = \mathfrak{p}_n R[x] \subsetneq \mathfrak{p}_n R[x] + \langle x \rangle$$

so $\dim(S) \geq \dim(R) + 1$.

4. Going down for normal domains gives us that $\text{ht}(\mathfrak{q}) \geq \text{ht}(\mathfrak{p})$. To see that $\text{ht}(\mathfrak{q}) \leq \text{ht}(\mathfrak{p})$ we consider a longest chain that ends at $\mathfrak{q}$ in B . Restricting it to A gives us a chain that ends at $\mathfrak{p}$. Moreover we know all inclusions in the chain are strict by Incomparability. Hence, $\text{ht}(\mathfrak{q}) = \text{ht}(\mathfrak{p})$.
5. Since $\mathbb{K}[y_1, \dots, y_d]$ is a polynomial ring over a field it is a UFD and hence is integrally closed and the extension $\mathbb{K}[y_1, \dots, y_d] \subseteq B$ is integral. Moreover note that any prime in $\mathbb{K}[y_1, \dots, y_d]$ has height at most d . Then by part 4 for any prime ideal $\mathfrak{q} \in B$ we have,

$$\text{ht}(\mathfrak{q}) = \text{ht}(\mathfrak{q} \cap \mathbb{K}[y_1, \dots, y_d]) \leq d$$

We now move from arbitrary integral extensions to finitely generated algebras over a field. The next theorem is one of the main structural results in commutative algebra. It says that a finitely generated algebra over a field is finite over a polynomial subring.

Theorem 0.4 (Noether Normalization). *Let $\mathbb{K}$ be a field and let A be an algebra-finite $\mathbb{K}$ -algebra. Then there exist algebraically independent elements $y_1, \dots, y_d \in A$ such that A is module-finite over the polynomial subring*

$$\mathbb{K}[y_1, \dots, y_d] \subseteq A.$$

If A is a domain, then $d = \text{tr.deg}_{\mathbb{K}} \text{Frac}(A)$.

The proof is an induction on the number of algebra generators. The key step is a change of variables trick: given a nonzero polynomial relation, we choose new variables so that one old variable satisfies a monic equation over the others.

Problem 11. The Monic Change of Variables Trick.

Let $0 \neq f \in \mathbb{K}[x_1, \dots, x_n]$. The goal of this exercise is to make a change of variables so that f becomes monic in one variable. Write

$$f = \sum_{\alpha} c_{\alpha} x_1^{\alpha_1} \cdots x_n^{\alpha_n},$$

where $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$, only finitely many c_{α} are nonzero, and $c_{\alpha} \in \mathbb{K}$. Choose an integer N larger than every exponent appearing in every monomial of f ; that is, $N > \alpha_i$ for every exponent vector α appearing in f and every i .

1. Define

$$w(\alpha) = \alpha_n + \alpha_1 N + \alpha_2 N^2 + \cdots + \alpha_{n-1} N^{n-1}.$$

Explain why this is the base- N expansion of an integer whose “digits” are $\alpha_n, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$

2. Since all these digits are strictly smaller than N , prove that if $w(\alpha) = w(\beta)$ then $\alpha = \beta$. Thus the weights $w(\alpha)$ are distinct for distinct monomials of f .
3. Introduce new variables

$$y_i = x_i - x_n^{N^i} \quad \text{for } 1 \leq i \leq n-1.$$

Equivalently, $x_i = y_i + x_n^{N^i}$. Substitute these expressions into a monomial

$$x_1^{\alpha_1} \cdots x_{n-1}^{\alpha_{n-1}} x_n^{\alpha_n}.$$

Show that the unique term obtained by choosing $x_n^{N^i}$ from every factor $(y_i + x_n^{N^i})^{\alpha_i}$ is $x_n^{w(\alpha)}$. All other terms contain at least one y_i and have strictly smaller degree in x_n .

4. Since the weights $w(\alpha)$ are distinct, there is a unique exponent vector $\alpha^{\max}$ appearing in f for which $w(\alpha)$ is largest. Show that, after the substitution, the highest power of x_n appearing in f is

$$c_{\alpha^{\max}} x_n^{w(\alpha^{\max})}.$$

5. Conclude that, after multiplying f by the nonzero scalar $c_{\alpha^{\max}}^{-1}$ we may regard f as a monic polynomial in x_n with coefficients in

$$\mathbb{K}[y_1, \dots, y_{n-1}].$$

6. Let $A = \mathbb{K}[x_1, \dots, x_n] / \langle f \rangle$. Prove that the image of x_n in A satisfies a monic polynomial with coefficients in the subalgebra generated by the images of

$$y_1, \dots, y_{n-1}.$$

Therefore the image of x_n is integral over this subalgebra.

7. Show A is generated as an algebra over $\mathbb{K}[y_1, \dots, y_{n-1}]$ by the single integral element x_n . Deduce that A is module-finite over $\mathbb{K}[y_1, \dots, y_{n-1}]$.

Solution.

1. Since $\alpha \in \mathbb{Z}$, and all $\alpha_i < N$ we can write a base- N expansion of the integer $(\alpha_n \alpha_1 \alpha_2 \dots \alpha_{n-1})_N$ in this manner.
2. Assume $w(\alpha) = w(\beta)$. Then $(\alpha_n \alpha_1 \alpha_2 \dots \alpha_{n-1})_N = (\beta_n \beta_1 \beta_2 \dots \beta_{n-1})_N \implies \alpha = \beta$.
3. We have $\prod_i^n x_i^{\alpha_i} = x_n^{\alpha_n} \prod_i^{n-1} (y_i + x_n^{N^i})^{\alpha_i}$. Picking just $x_n^{N^i}$ from every factor gives us,

$$x_n^{\alpha_n} \prod_i^{n-1} (x_n^{N^i})^{\alpha_i} = x_n^{\alpha_n} \prod_i^{n-1} x_n^{\alpha_i N^i} = x_n^{\alpha_n + \alpha_1 N^1 + \dots + \alpha_{n-1} N^{n-1}} = x_n^{w(\alpha)}$$

By binomial expansion we know that all other terms contain at least one y_i and hence have strictly smaller degree in x_n .

4. Thus after substitution, the leading term contributed by each monomial $c_\alpha x_1^{\alpha_1} \dots x_n^{\alpha_n}$ in f is $c_\alpha x_n^{w(\alpha)}$. From part 2 we know that distinct α 's have distinct $w(\alpha)$'s, and thus no two monomials of f contribute leading terms of the same order. Therefore, there exists some $\alpha^{\max}$ whose weight $w(\alpha^{\max})$ is the largest, and thus the highest power of x_n in f is $c_{\alpha^{\max}} x_n^{w(\alpha^{\max})}$.
5. Since $c_\alpha \in \mathbb{K}$, we can multiply the term in the previous part by $c_{\alpha^{\max}}^{-1}$ to ensure the coefficient is one, thus making the polynomial f monic in x_n with coefficients in $\mathbb{K}[y_1, \dots, y_{n-1}]$.
6. $x_n \in \mathbb{K}[x_1, \dots, x_n]$ satisfies the monic polynomial

$$c_{\alpha^{\max}}^{-1} \cdot f = x_n^{w(\alpha^{\max})} + g_{d-1}(y_1, \dots, y_{n-1}) x_n^{w(\alpha^{\max})-1} + \dots + g_0(y_1, \dots, y_{n-1}) = 0$$

with coefficients in $\mathbb{K}[y_1, \dots, y_{n-1}]$. In A we get,

$$\overline{x_n} = \overline{c_{\alpha^{\max}}^{-1} \cdot f} = \overline{x_n^{w(\alpha^{\max})}} + \overline{g_{d-1} x_n^{w(\alpha^{\max})-1}} + \dots + \overline{g_0} = 0$$

where each $\overline{g_i}$ is the image of $g_i(y_1, \dots, y_{n-1})$ in A . Hence $\overline{x_n}$ is integral over the subalgebra generated by the image of $y_1, \dots, y_{n-1}$ in $\mathbb{K}[x_1, \dots, x_n] / \langle f \rangle$.

7. Note that there is an isomorphism $\mathbb{K}[y_1, \dots, y_{n-1}][x_n] \cong \mathbb{K}[x_1, \dots, x_n]$ given by the maps,

$$\phi : x_i \mapsto y_i + x_n^{N_i}, \quad x_n \mapsto x_n$$

$$\psi : y_i \mapsto x_i - x_n^{N_i}, \quad x_n \mapsto x_n$$

Now we show that the map $\pi : \mathbb{K}[y_1, \dots, y_{n-1}] \rightarrow A$ is injective. Assume for contradiction $p(y_1, \dots, y_{n-1}) \in \ker(\pi)$. That is, $p(y_1, \dots, y_{n-1}) \in \langle f \rangle$ or $p(y_1, \dots, y_{n-1}) = g \cdot f$ for some $g \in \mathbb{K}[x_1, \dots, x_n]$. But $x_n - \deg(p) = 0$ and $x_n - \deg(g \cdot f) \geq w(\alpha^{\max})$. But this is a contradiction. Hence π must be injective.

From the isomorphism, we know that every element of A can be represented by $h \in \mathbb{K}[x_1, \dots, x_n] = \mathbb{K}[y_1, \dots, y_{n-1}][x_n]$. Since f is monic we get,

$$h = f \cdot q + r, \quad x_n - \deg(r) \leq w(\alpha^{\max})$$

Since $f \equiv 0$ in A , we get $\bar{h} = \bar{r}$ where $x_n - \deg(r) \leq w(\alpha^{\max})$ with coefficients in $\mathbb{K}[y_1, \dots, y_{n-1}]$. Thus A is spanned by $1, x_n, \dots, x_n^{w(\alpha^{\max})-1}$ and hence is module-finite.

Problem 12. Proof of Noether Normalization.

Let $A = \mathbb{K}[x_1, \dots, x_n]/I$ be an algebra-finite $\mathbb{K}$ -algebra.

1. If $I = \langle 0 \rangle$, prove the theorem.
2. If $I \neq \langle 0 \rangle$, choose $0 \neq f \in I$. Use the previous exercise to make a change of variables so that A is module-finite over the image of a polynomial ring in $n - 1$ variables.
3. Let A_1 denote this image subalgebra. Explain why A_1 is algebra-finite over $\mathbb{K}$ using at most $n - 1$ algebra generators.
4. Apply induction to A_1 to find a polynomial subring $\mathbb{K}[y_1, \dots, y_d] \subseteq A_1$ such that A_1 is module-finite over $\mathbb{K}[y_1, \dots, y_d]$.
5. Prove that A is module-finite over $\mathbb{K}[y_1, \dots, y_d]$.
6. Explain why the elements $y_1, \dots, y_d$ are algebraically independent over $\mathbb{K}$.

Solution.

1. If $I = \langle 0 \rangle$, then the x_i are algebraically independent and $A = \mathbb{K}[x_1, \dots, x_n]$ is itself a polynomial ring and hence is trivially module-finite over itself.
2. Assume $I \neq \langle 0 \rangle$. Now let f be a non-zero polynomial in I . By Problem 11 we know that $\mathbb{K}[x_1, \dots, x_n]/\langle f \rangle$ is module-finite over $\mathbb{K}[y_1, \dots, y_{n-1}]$ for some algebraically independent $y_1, \dots, y_{n-1}$.
Since $f \in I$ we have $\langle f \rangle \subseteq I$ and we have the quotient map $\mathbb{K}[x_1, \dots, x_n]/\langle f \rangle \rightarrow A$, and hence A is module-finite over the image of $\mathbb{K}[y_1, \dots, y_{n-1}]$ in A .
3. Since A_1 is the image of a polynomial ring in $n - 1$ variables, it has at most $n - 1$ algebra-generators over $\mathbb{K}$.
4. By part 3, A_1 has at most $n - 1$ generators, so by the induction hypothesis there exists some $d \leq n - 1$ and algebraically independent elements $y_1, \dots, y_d$ such that A_1 is module finite over $\mathbb{K}[y_1, \dots, y_d]$.
5. Since A is module-finite over A_1 and A_1 is module-finite over $\mathbb{K}[y_1, \dots, y_d]$, A is module-finite over $\mathbb{K}[y_1, \dots, y_d]$ (Worksheet 7.1, Problem 7).
6. $\mathbb{K}[y_1, \dots, y_d]$ is a polynomial ring and thus the y_i are algebraically independent over $\mathbb{K}$.

Problem 13. Finding Normalizations by Hand.

For each algebra below, find a polynomial subring over which the algebra is module-finite. When possible, give explicit module generators.

- | | |
|--|--|
| 1. $\mathbb{K}[t^2, t^3]$. | 4. $\mathbb{K}[x, y, z] / \langle z^2 - xy \rangle$. |
| 2. $\mathbb{K}[x, y] / \langle y^2 - x^3 \rangle$. | 5. $\mathbb{K}[x^2, xy, y^2] \subseteq \mathbb{K}[x, y]$. |
| 3. $\mathbb{K}[x, y] / \langle xy \rangle$. Hint: Try $x + y$. | 6. $\mathbb{K}[s^2, s^3, t]$. |

Solution.

1. $\mathbb{K}[t^2, t^3]$ is module-finite over $\mathbb{K}[t^3 - t^2]$ with generators $\{1, t^2, t^4\}$.
2. $\mathbb{K}[x, y] / \langle y^2 - x^3 \rangle$ is module-finite over $\mathbb{K}[x]$ with generators $\{1, y\}$.
3. $\mathbb{K}[x, y] / \langle xy \rangle$ is module-finite over $\mathbb{K}[x + y]$ with generators $\{1, x, y\}$.
4. $\mathbb{K}[x, y, z] / \langle z^2 - xy \rangle$ is module-finite over $\mathbb{K}[x + y, z]$ with generators $\{1, x\}$.
5. $\mathbb{K}[x^2, xy, y^2] \subseteq \mathbb{K}[x, y]$ is module-finite over $\mathbb{K}[x^2, y^2]$ with generators $\{1, xy\}$.
6. $\mathbb{K}[s^2, s^3, t]$ is module-finite over $\mathbb{K}[s^3 - s^2, t]$ with generators $\{1, s^2, s^4\}$.

Problem 14. Zariski's Lemma from Noether Normalization.

Let A be an algebra-finite $\mathbb{K}$ -algebra that is also a field.

1. By Noether Normalization, choose $\mathbb{K}[y_1, \dots, y_d] \subseteq A$ such that A is module-finite over $\mathbb{K}[y_1, \dots, y_d]$. Explain why this extension is integral.
2. Use the earlier fact that if B is a field integral over a domain A_0 , then A_0 is a field, to prove that $\mathbb{K}[y_1, \dots, y_d]$ is a field.
3. Prove that $d = 0$.
4. Conclude that A is a finite-dimensional vector space over $\mathbb{K}$. Equivalently, every field that is algebra-finite over $\mathbb{K}$ is algebraic and finite over $\mathbb{K}$.
5. Deduce the weak Nullstellensatz: if $\mathbb{K}$ is algebraically closed and $\mathfrak{m} \subset \mathbb{K}[x_1, \dots, x_n]$ is maximal, then

$$\mathfrak{m} = \langle x_1 - a_1, \dots, x_n - a_n \rangle$$

for some $a_1, \dots, a_n \in \mathbb{K}$.

Solution.

1. Module-finiteness implies integrality hence A is integral over $\mathbb{K}[y_1, \dots, y_d]$.
2. We have an integral extension $\mathbb{K}[y_1, \dots, y_d] \rightarrow A$, and we know that $\mathbb{K}[y_1, \dots, y_d]$ is a polynomial ring over a field and hence a UFD. Then by Problem 1.2, since A is a field, $\mathbb{K}[y_1, \dots, y_d]$ is also a field.
3. Assume $d \neq 0$. Then $y_1 \in \mathbb{K}[y_1, \dots, y_d]$ doesn't have an inverse. A contradiction. Hence d must be 0.
4. Thus A is module-finite over the field $\mathbb{K}$, i.e. is a finite-dimensional vector space over $\mathbb{K}$, i.e. every element of A is algebraic over $\mathbb{K}$. So A is an algebra-finite over $\mathbb{K}$.
5. Let $\mathfrak{m} \subset \mathbb{K}[x_1, \dots, x_n]$ be maximal. Then $\mathbb{K}[x_1, \dots, x_n]/\mathfrak{m}$ is a field. By Noether Normalization $\mathbb{K}[x_1, \dots, x_n]/\mathfrak{m}$ is algebra-finite over $\mathbb{K}$ and hence a finite algebraic extension of $\mathbb{K}$. Since $\mathbb{K}$ is algebraically closed it has no finite proper extensions and hence $\mathbb{K}[x_1, \dots, x_n]/\mathfrak{m} \cong \mathbb{K}$. Then $x_i - a_i \in \mathfrak{m}$ and hence $\langle x_1 - a_1, \dots, x_n - a_n \rangle = \mathfrak{m}$.

We finish by extracting consequences for dimension. Noether Normalization reduces the dimension theory of affine algebras to the dimension theory of polynomial rings, while Lying Over, Going Up, and Incomparability tell us that finite integral extensions preserve lengths of prime chains.

Theorem 0.5 (Dimension of Affine Domains). *Let A be an algebra-finite $\mathbb{K}$ -algebra that is a domain. Then*

$$\dim(A) = \text{tr.deg}_{\mathbb{K}} \text{Frac}(A).$$

Equivalently, if $\mathbb{K}[y_1, \dots, y_d] \subseteq A$ is a Noether normalization, then $\dim(A) = d$.

Problem 15. Dimension of Affine Domains.

Let A be an algebra-finite $\mathbb{K}$ -algebra that is a domain.

1. Let $\mathbb{K}[y_1, \dots, y_d] \subseteq A$ be a Noether normalization. Use dimension preservation for integral extensions to prove $\dim(A) = d$.
2. Show that $\text{Frac}(A)$ is algebraic over $\mathbb{K}(y_1, \dots, y_d)$. Hint: First prove that every element of A is algebraic over $\mathbb{K}(y_1, \dots, y_d)$.
3. Deduce that $\text{tr.deg}_{\mathbb{K}} \text{Frac}(A) = d$.
4. Conversely, explain why d is determined by A and not by the choice of Noether normalization.

Solution.

1. A is integral over $\mathbb{K}[y_1, \dots, y_d]$. We know that $\dim(\mathbb{K}[y_1, \dots, y_d]) = d$ and by laying over and going up $\dim(A) \geq d$. Let

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \dots \subsetneq \mathfrak{q}_n$$

be a prime ideal chain in A . Then by Incomparability we get the chain

$$\mathfrak{q}_0 \cap \mathbb{K}[y_1, \dots, y_d] \subsetneq \mathfrak{q}_1 \cap \mathbb{K}[y_1, \dots, y_d] \subsetneq \dots \subsetneq \mathfrak{q}_n \cap \mathbb{K}[y_1, \dots, y_d]$$

Hence every length in A corresponds to a chain of equal length in $\mathbb{K}[y_1, \dots, y_d]$ and $\dim(A) = d$.

2. Since A is integral over $\mathbb{K}[y_1, \dots, y_d]$ every element of A satisfies some monic polynomial with coefficients in $\mathbb{K}[y_1, \dots, y_d]$, i.e. every element in A is algebraic over $\mathbb{K}[y_1, \dots, y_d]$. Since $\mathbb{K}[y_1, \dots, y_d] \subseteq \mathbb{K}(y_1, \dots, y_d)$, every element of A is algebraic over it.

Now note that elements of $\text{Frac}(A)$ are of the form $\frac{a}{b}$ where $a, b \in A$. Since both a, b are algebraic over $\mathbb{K}(y_1, \dots, y_d)$ and algebraic extensions are closed under field operations $\frac{a}{b}$ is also algebraic over $\mathbb{K}(y_1, \dots, y_d)$.

3. We know that $\text{tr.deg}_{\mathbb{K}} \text{Frac}(A) \geq d$ since $y_1, \dots, y_d$ are algebraically independent over $\mathbb{K}$. By part 2, since $\text{Frac}(A)$ is algebraic over $\mathbb{K}(y_1, \dots, y_d)$, $\text{tr.deg}_{\mathbb{K}} \text{Frac}(A) \leq d$ since every element except $y_1, \dots, y_d$ are algebraic over $\mathbb{K}(y_1, \dots, y_d)$. Thus, $\text{tr.deg}_{\mathbb{K}} \text{Frac}(A) = d$.
4. Because $\text{tr.deg}_{\mathbb{K}} \text{Frac}(A) = d$, which depends only on A and $\mathbb{K}$, not on any choices made in the Noether normalization.

Problem 16. Computations Using Noether Normalization.

Compute the Krull dimension of each ring. In each case, try to exhibit a finite extension from a polynomial subring or reduce to irreducible components.

- | | |
|---|--|
| 1. $\mathbb{K}[t^2, t^3]$. | 5. $\mathbb{K}[x, y, z] / \langle xz, yz \rangle$. |
| 2. $\mathbb{K}[s^2, s^3, t]$. | 6. $\mathbb{K}[x_1, \dots, x_n] / \langle x_1, \dots, x_r \rangle$. |
| 3. $\mathbb{K}[x, y, z] / \langle z^2 - xy \rangle$. | 7. $\mathbb{Z}[x] / \langle x^2 + 1 \rangle$. |
| 4. $\mathbb{K}[x, y, z] / \langle xy, z \rangle$. | 8. $\mathbb{K}[x^2, xy, y^2]$. |

Solution.

1. Since $\mathbb{K}[t^2, t^3]$ is module-finite over $\mathbb{K}[t^3 - t^2]$ we have $\dim(\mathbb{K}[t^2, t^3]) = \dim(\mathbb{K}[t^3 - t^2]) = 1$.
2. Since $\mathbb{K}[s^2, s^3, t]$ is module-finite over $\mathbb{K}[s^3 - s^2, t]$ we have $\dim(\mathbb{K}[s^2, s^3, t]) = \dim(\mathbb{K}[s^3 - s^2, t]) = 2$.
3. Since $\mathbb{K}[x, y, z] / \langle z^2 - xy \rangle$ is module-finite over $\mathbb{K}[x + y, z]$ we have $\dim(\mathbb{K}[x, y, z] / \langle z^2 - xy \rangle) = \dim(\mathbb{K}[x + y, z]) = 2$.
4. Since $\mathbb{K}[x, y, z] / \langle xy, z \rangle$ is module-finite over $\mathbb{K}[x + y]$ we have $\dim(\mathbb{K}[x, y, z] / \langle xy, z \rangle) = \dim(\mathbb{K}[x + y]) = 1$.
5. $\mathbb{K}[x, y, z] / \langle xz, yz \rangle = \mathbb{K}[x, y, z] / \langle z \rangle \cap \langle x, y \rangle$. Then,

$$\mathbb{K}[x, y, z] / \langle z \rangle \cong \mathbb{K}[x, y] \implies \dim(\mathbb{K}[x, y, z] / \langle xz, yz \rangle) = 2$$

or

$$\mathbb{K}[x, y, z] / \langle x, y \rangle \cong \mathbb{K}[z] \implies \dim(\mathbb{K}[x, y, z] / \langle xz, yz \rangle) = 1$$

Since the Krull dimension is the sup, $\dim(\mathbb{K}[x, y, z] / \langle xz, yz \rangle) = 2$.

6. Since $\mathbb{K}[x_1, \dots, x_n] / \langle x_1, \dots, x_r \rangle$ is module-finite over $\mathbb{K}[x_{r+1}, \dots, x_n]$ we have $\dim(\mathbb{K}[x_1, \dots, x_n] / \langle x_1, \dots, x_r \rangle) = \dim(\mathbb{K}[x_{r+1}, \dots, x_n]) = n - r$.
7. Since $\mathbb{Z}[x] / \langle x^2 + 1 \rangle$ is module-finite over $\mathbb{Z}$ we have $\dim(\mathbb{Z}[x] / \langle x^2 + 1 \rangle) = \dim(\mathbb{Z}) = 1$.
8. Since $\mathbb{K}[x^2, xy, y^2]$ is module-finite over $\mathbb{K}[x^2, y^2]$ we have $\dim(\mathbb{K}[x^2, xy, y^2]) = \dim(\mathbb{K}[x^2, y^2]) = 2$.

Problem 17. Finite Maps to Affine Space.

Let A be an algebra-finite $\mathbb{K}$ -algebra and let $P = \mathbb{K}[y_1, \dots, y_d] \subseteq A$ be a Noether normalization.

1. Show that the induced map $\text{Spec}(A) \longrightarrow \text{Spec}(P) \cong \mathbb{A}_{\mathbb{K}}^d$ is surjective.
2. For $\mathfrak{p} \in \text{Spec}(P)$, prove that the fibre ring $\kappa(\mathfrak{p}) \otimes_P A$ is finite-dimensional over $\kappa(\mathfrak{p})$.
3. Assume $\mathbb{K}$ is algebraically closed and $\mathfrak{p} = \langle y_1 - a_1, \dots, y_d - a_d \rangle$ is a closed point of $\mathbb{A}_{\mathbb{K}}^d$. Show that the closed points in the fibre over $\mathfrak{p}$ correspond to maximal ideals of the finite-dimensional $\mathbb{K}$ -algebra $A/\mathfrak{p}A$.
4. Prove that a finite-dimensional algebra over a field has only finitely many maximal ideals. Hint: Distinct maximal ideals are comaximal, and the Chinese Remainder Theorem gives a surjection onto a product of fields.
5. Explain the geometric slogan: Noether normalization gives a finite-to-one projection from an affine algebraic set to affine space of the same dimension.

Solution.

1. By Laying over every ideal $\mathfrak{p} \in \text{Spec}(P)$ corresponds to an ideal $\mathfrak{q} \in \text{Spec}(A)$ such that $\mathfrak{q} \cap P = \mathfrak{p}$. Hence the map $\text{Spec}(A) \rightarrow \text{Spec}(P)$ is surjective.

2. We proved this in Problem 8.

3. Since $\mathbb{K}$ is algebraically closed, $\mathfrak{p}$ is maximal in P and $\kappa(\mathfrak{p}) = P/\mathfrak{p} \cong \mathbb{K}$. So the fibre ring is

$$\kappa(\mathfrak{p}) \otimes_P A = (P/\mathfrak{p}) \otimes_P A \cong A/\mathfrak{p}A$$

The closed points of the fiber are the maximal ideals of this fiber ring, which are exactly the maximal ideals of A containing $\mathfrak{p}A$ i.e. the maximal ideals of $A/\mathfrak{p}A$.

4. Let $\mathfrak{m}_1, \dots, \mathfrak{m}_r$ be distinct maximal ideals of $A/\mathfrak{p}A$. Since they are distinct maximal ideals, they are pairwise comaximal and by the CRT, there is a surjection

$$A/\mathfrak{p}A \twoheadrightarrow \prod_{i=1}^r A/\mathfrak{m}_i$$

Each $\mathfrak{m}_i$ is a maximal ideal of $A/\mathfrak{p}A$, which is a finite-dimensional $\mathbb{K}$ -algebra and hence algebra-finite over $\mathbb{K}$. By the Weak Nullstellensatz, $A/\mathfrak{m}_i \cong \mathbb{K}$ for each i . Thus we have a surjection

$$A/\mathfrak{p}A \twoheadrightarrow \prod_{i=1}^r \mathbb{K}$$

which gives

$$\dim_{\mathbb{K}}(A/\mathfrak{p}A) \geq \dim_{\mathbb{K}} \prod_{i=1}^r \mathbb{K} = r$$

Since $A/\mathfrak{p}A$ is finite-dimensional over $\mathbb{K}$ by part (2) and $r \leq \dim_{\mathbb{K}}(A/\mathfrak{p}A)$, $A/\mathfrak{p}A$ has only finitely many maximal ideals.

5. Noether normalization gives a surjection $\text{Spec}(A) \twoheadrightarrow A_{\mathbb{K}}^d$ where $d = \dim(A)$. Parts (3) and (4) say that each fiber over a closed point is finite and nonempty. So geometrically, every point of $A_{\mathbb{K}}^d$ has finitely many points of $\text{Spec}(A)$ above it and hence Noether normalization gives a finite-to-one projection from an affine algebraic set to affine space of the same dimension.